Open Source Audit Report

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Project Goal :- Open Source Audit Report

On Python

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1. Introduction

Open Source Software (OSS) plays a vital role in modern computing by promoting collaboration, transparency, and innovation. This project focuses on auditing **Python**, one of the most widely used open-source programming languages.

The objective of this audit is to analyze system-level interactions, package management, disk usage, log monitoring, and user interaction using shell scripting in a Linux environment.

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2. Objectives

- To understand the working of open-source systems.

- To develop practical knowledge of shell scripting.
- To analyze system information and resources.
- To interact with Linux utilities for auditing purposes.
- To explore Python as an open-source software ecosystem.

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3. Overview of Python

Python is a high-level, interpreted programming language known for its simplicity and readability. It is widely used in:

- Web development
- Data Science & AI
- Automation

- Cybersecurity

Being open source, Python allows developers worldwide to contribute and improve its ecosystem.

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4. System Environment

- **Operating System**: Linux-based system
- **Shell Used**: Bash (/bin/bash)
- **Tools Used**:
  - awk, grep, cut, tail
  - dpkg / rpm
  - du, ls

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5. Project Components (Shell Scripts)

5.1 System Identity Script (sys_identity.sh)

- Displays:
  - Kernel version
  - OS distribution
  - Logged-in user
  - System uptime
- Uses:
  - Command substitution
  - System variables

👉 Purpose: Provides basic system audit information.

5.2 Package Inspector (package_inspector.sh)

- Checks whether python3 is installed.
- Uses:
  - dpkg or rpm
  - case statements

👉 Purpose: Verifies software installation and demonstrates conditional logic.

5.3 Disk Auditor (disk_auditor.sh)

- Iterates through directories.
- Displays:
  - Directory size
  - Permissions
- Uses:
  - Arrays

- du, ls -ld, awk

👉 Purpose: Monitors disk usage and access control.

5.4 Log Analyzer (log_analyzer.sh)

- Reads log files line by line.
- Counts occurrences of keywords (e.g., "error").
- Displays last 5 matches.

👉 Purpose: Helps in identifying system issues and debugging.

5.5 Manifesto Generator (manifesto.sh)

- Takes user input interactively.

- Generates a custom Open Source philosophy file.

👉 Purpose: Demonstrates user interaction and file creation.

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6. Execution Process

1. Open terminal in project directory
2. Grant execution permission:

```
chmod +x *.sh
```

3. Run scripts:

```
./script_name.sh
```

Example:

```
./sys_identity.sh
```

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7. Key Learnings

- Understanding of Linux command-line tools
- Hands-on experience with Bash scripting
- Importance of open-source contributions
- System-level auditing techniques
- File handling and automation

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8. Advantages of Open Source (Python)

- Free to use and distribute
- Large community support
- Highly customizable
- Secure due to transparency

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9. Limitations

- Requires technical knowledge
- Dependency management issues
- Performance limitations in some cases

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10. Conclusion

This project successfully demonstrates how open-source tools like Python can be audited using shell scripting. It enhances practical knowledge of Linux systems, scripting, and software analysis.

The scripts developed provide insights into system behavior, resource usage, and log monitoring, making this project highly relevant for real-world

applications in system administration and cybersecurity.

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11. Future Scope

- Integration with automated dashboards
- Advanced log analysis using AI
- Security vulnerability scanning
- Cloud-based system auditing

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12. References

- Official Python Documentation
- Linux Manual Pages (man commands)
- Open Source Community Resources